MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 11732350 RETICLE PROJECTOR AND ELEVATION SERVO DRIVE

This specification is approved for use by the US Army Armament Research and Development Command (ARRADCOM) and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements and quality assurance provisions for the Circuit Card Assembly: 11732350 Reticle Projector and Elevation Servo Drive which is the A3 component of the M21 Ballistic Computer Unit, 11732400.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-13926	Calibration System Requirements
MIL-I-45607	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-STD-45662	Inspection Equipment, Acquisition, Maintenance and Disposition of

FSC 1220

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 202640-22T19:28Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-810	Environmental Test Methods

DRAWINGS

ARRADCOM

11732350	Circuit Card Assembly: 11732354 Reticle Projector and Elevation Servo Drives

Inspection Equipment

11750202	Automatic Card and Unit, Test Set
11750354	Adapter, Computer Board - A3
11821457	Fixture Shock, Vertical and Horizontal Plane
11821458	Shock Adapter, Plate Vib Horz - Vert Plane/ Shock -Z
11821500	Vibration/Shock, Test Fixture - Computer
11821520	Automatic Test Assembly Central Processor Unit
11821854	Test Fixture, Auto Card, Test Set, Computer FAT

Packaging Data Sheet

P11732350	Packaging of Circuit Card Assembly: 11732350

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:28Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Fabrication. The reticle projector and elevation servo drive circuit card assembly, herein referred to as the assembly, shall be manufactured in accordance with Drawing 11732350 and drawings pertaining thereto and when assembled, shall meet the requirements of this specification. (See 4.5.1)

3.1.1 Signals. The assembly shall provide the following signals:

Signal Designation	Signal Nomenclature
a. D SERVO MOTOR	Reticle projector motor driver
b. E MOTOR 1	Output unit motor drive one
c. E MOTOR 2	Output unit motor drive two

3.1.2 General specifications. The following provisions of MIL-F-13926 apply: (See 4.5.1)

a. Order of procedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Part identification and marking
e. Workmanship

3.1.3 Ambient conditions. Standard ambient conditions shall be as follows:

a. Temperature	$73^{\circ} \pm 18^{\circ}F$
b. Relative humidity	50 percent $\pm$ 30 percent
c. Atmospheric pressure	28.5 + 2.0 $\overline{-}$ 3.0 in. Hg.

3.2 First article. When specified (See 6.2), the contractor shall furnish sample units for first article inspection and approval (See 4.4 and 6.2).

3.3 Performance. Unless otherwise specified, the assembly shall meet the performance requirements specified herein under standard ambient conditions of 3.1.3.

3.3.1 D Servo Motor output. With the load (item 1.3 of table I) connected, the power sources (items 2.1, 2.2 and 2.3 of table I) applied and the signal source (item 3.1 of table I) applied, the nominal output measured differentially between V1 and V2 (see figure 1) shall be described by the following transfer function:

$$(V1 - V2) = 19.35 * V_{28} - 0.42$$

Source: https://assist.dla.mil -- Downloaded: 202640-22T19:28Z
Check the source to verify that this is the currenversion before use.

(V_1 - V_2) shall be 180 ± 10 degrees out of phase with reference to the signal source. The tolerance of the output shall be not greater than than value of the following equation:

$$\Delta (V1 - V2) = 0.05 * (V1 - V2) + 0.10$$

Application of the transfer function and tolerance equation shall be valid within the following limit:

$$|(V1 - V2)| < 27 \text{ vrms}$$ (See 4.6.2.1)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:28Z
Check the source to verify that this is the current version before use.

TABLE 1. Loads, power and signal sources.

Item	Conditions	Characteristics	Connections
1.0	Loads		Connected between the following pins when specified:
1.1	Resistor	4.9 ohms ± 5% 100 watts	P1-2A and P1-23A
1.2	Resistor	51.1 ohms ± 5% 10 watts	P1-2A and P1-23A
1.3	Motor Simulator	See figure 1	P1-4A and P1-3A
2.0	Power sources		Connected between the following pins:
2.1	24.0 vdc (24V COMP 3)	Tolerance: ± 1.0 vdc Ripple: 500 mv p-p max	P1-19A, P1-19B and P1-24A, P1-24B (rtn)
2.2	15.000 vdc (+15V)	Tolerance: ± 0.150 vdc Ripple: 20 mv p-p max	P1-6A, P1-6B and P1-13A, P1-13B (rtn)
2.3	-15.000 vdc (-15V)	Tolerance: ± 0.150 vdc Ripple: 20 mv p-p max	P1-21A, P1-21B and P1-13A, P1-13B (rtn)
3.0	Signal sources		Connected between the following pins:
3.1	400 hz sine wave (D PWR AMP)	Adjustable: 0.0 to -10.0 vrms Tolerance: ± 4 hz	P1-2B and P1-13A (rtn)
3.2	-DC voltage	Adjustable: 0.0 to -10.0 vdc Ripple: 10 mv p-p max	
	(E PWR AMP 1) or (E PWR AMP 2)		P1-11A and P1-8A (rtn) or P1-5A and P1-8A (rtn)
4.0	Interconnections		Connected between the following pins:
4.1		Jumper wire	P1-24A, P1-24B and P1-8A, P1-8B
4.2		Jumper wire	P1-13A, P1-13B and P1-1A, P1-1B

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:28Z
Check the source to verify that this is the current version before use.

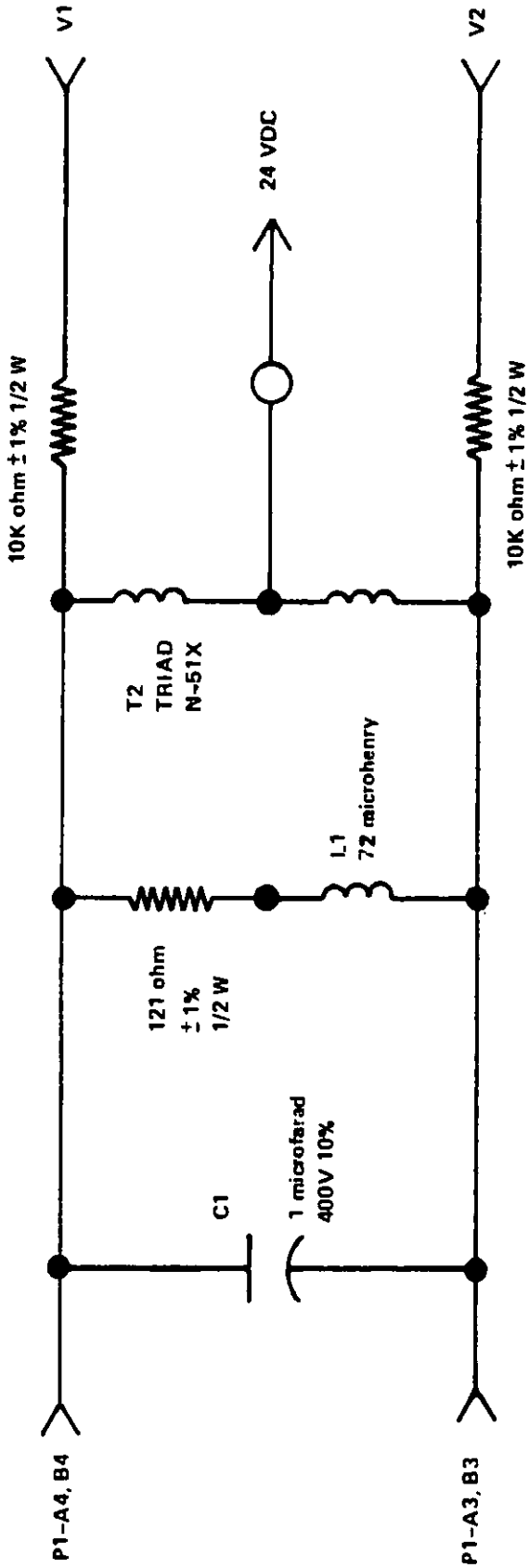


FIGURE 1. Motor simulator.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:28Z
Check the source to verify that this is the current version before use.

3.3.2 E MOTOR 1 and E MOTOR 2 outputs. With the load (item 1.1 or item 1.2 of table I) connected (see table II), the power sources (items 2.1, 2.2 and 2.3 of table I) applied and the signal source (item 3.2 of table I) applied as required to either P1-11A or P1-5A (see table II), the output voltage shall be as specified for each of the eight items of table II when the corresponding input and load conditions are established. (See 4.6.2.2)

TABLE II. E MOTOR 1 and E MOTOR 2 outputs.

Item	Input Conditions 1/		Load 2/	Output voltage (vdc) 3/
	P1-11A	P1-5A		
1	-7	Gnd	Max	20.00 ± 2.00
2	-4.6	Gnd	Max	13.50 ± 1.35
3	-7	Gnd	Min	21.50 ± 2.15
4	-4.6	Gnd	Min	15.00 ± 1.50
5	Gnd	-7	Max	-20.00 ± 2.00
6	Gnd	-4.6	Max	-13.50 ± 1.35
7	Gnd	-7	Min	-21.50 ± 2.15
8	Gnd	-4.6	Min	-15.00 ± 1.50

1/ -7 = -7.00 ± 0.10 vdc (item 3.2 of table I)
-4.6 = -4.60 ± 0.10 vdc (item 3.2 of table I)
Gnd = interconnection with P1-24A

2/ Max = item 1.1 of table I connected
Min = item 1.2 of table I connected

3/ Measured at P1-2A with respect to P1-23A

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:28Z
Check the source to verify that this is the current version before use.

3.4 Environmental.

3.4.1 Shock. The assembly shall be capable of operating as specified herein after exposure under the conditions of 3.1.3 to three half sine wave shock pulses of 40 ± 4 gravity unit (g) for a duration of 18 ± 3 milliseconds (msec) applied in each direction along three mutually perpendicular axes. In addition, the assembly shall be capable of withstanding three half sine wave shock pulses of 100.0 ± 10.0 g for a duration of 1.5 ± 0.2 msec applied in each direction along three mutually perpendicular axes. (See 4.6.3.1)

3.4.2 Vibration. The assembly shall be capable of operating as specified herein after exposure under the conditions of 3.1.3 to the vibration profile of figure 2. Duration of exposure shall be not less than 80 minutes in each of three mutually perpendicular axes. (See 4.6.3.2)

3.4.3 Operating temperature. The assembly shall be capable of operating as specified herein over the operating temperature range of -25°F to +140°F. (See 4.6.3.3, 4.6.3.4 and 4.6.3.5)

3.4.4 Storage temperature. The assembly shall be capable of operating as specified herein after exposure to storage temperatures ranging from -65°F to 160°F. (See 4.6.3.3, 4.6.3.4 and 4.6.3.5).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:28Z
Check the source to verify that this is the current version before use.

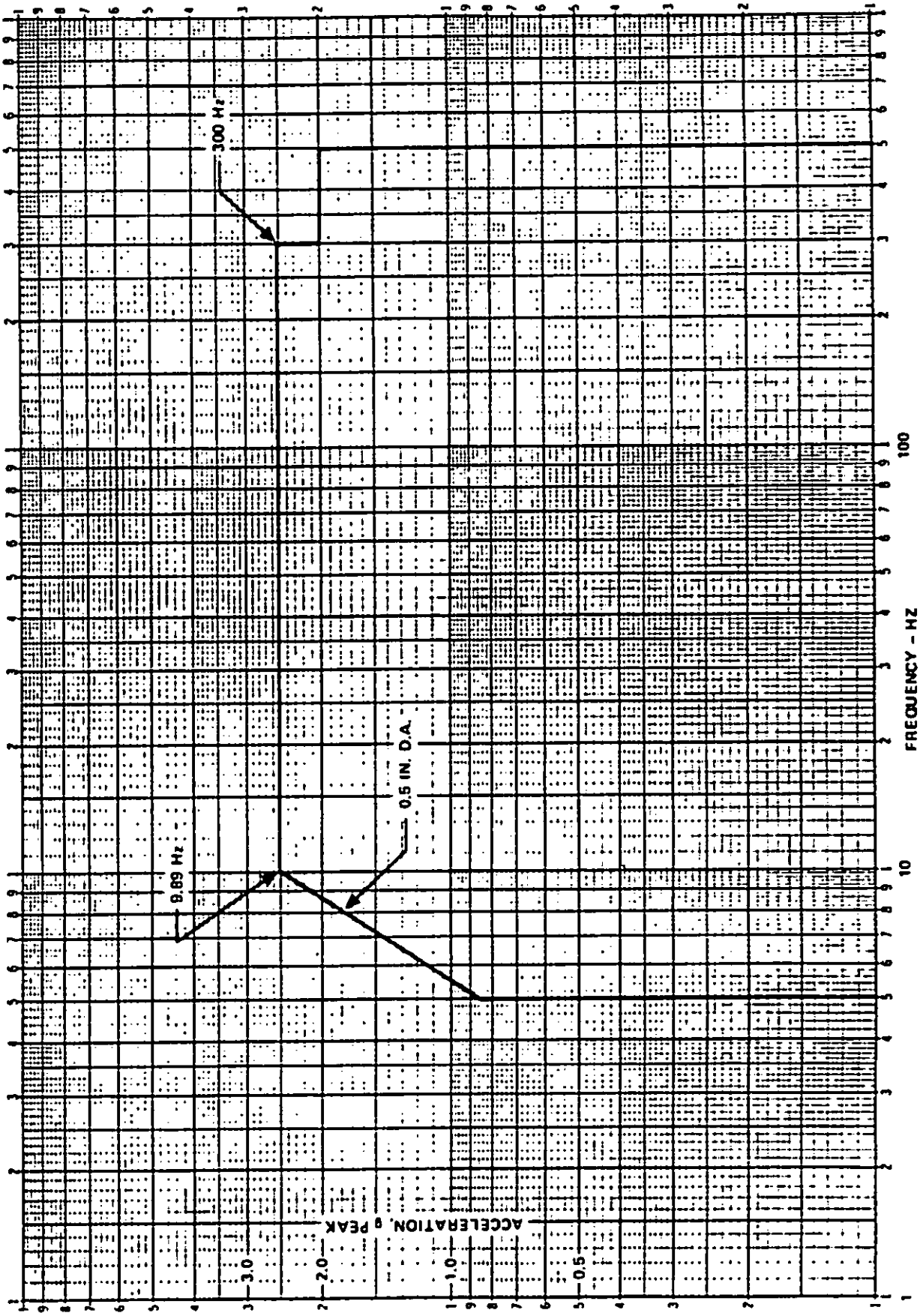


FIGURE 2. Vibration profile.

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:28Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilties suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.4).
2. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the quality assurance provisions of MIL-F-13926 and the conditions of 3.1.3.

4.4 First article. The requirements for first article approval and the designation of responsibility for first article inspection to either the Government or the contractor shall be specified in the contract.

4.4.1 Sample. The first article sample shall be three assemblies selected at random by the Government representative from the first fifteen production assemblies.

4.4.2 Inspection. The sample shall be subjected to all the inspections specified in tables III, IV and V.

4.4.3 Failure. Failure of any assembly to meet any requirement shall be cause for refusal to grant first article approval. The Government reserves the right to terminate first article inspection upon any failure of any assembly to comply with any stated requirement.

4.4.4 Responsibility. The contractor, whether or not responsible, shall inspect the sample for conformance to all contractual requirements and shall submit a record of this inspection with the sample and certification of conformance for materials. The Government reserves the right to witness the contractor's inspection.

Source: https://assist.dla.mil -- Downloaded: 202640-22T19:28Z
Check the source to verify that this is the current version before use.

TABLE III. Performance tests.

Item	Characteristic	Requirement	Test Procedures
101 102	D SERVO MOTOR output E MOTOR 1 and E MOTOR 2 outputs	3.3.1 3.3.2	4.6.2.1 4.6.2.2

TABLE IV. General tests.

Item	Characteristic	Requirement	Test Procedure
301	Fabrication	3.1	Applicable drawings - Visual
302	General Specification	3.1.2	MIL-F-13926 - Visual

TABLE V. Environmental tests.

Item	Characteristic	Requirement	Test Procedures
303	Shock	3.4.1	4.6.3.1
304	Vibration	3.4.2	4.6.3.2
305	High Temperature	3.4.3 and 3.4.4	4.6.3.3
306	Low Temperature	3.4.3 and 3.4.4	4.6.3.4
307	Temperature cycling	3.4.3 and 3.4.4	4.6.3.5

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:28Z
Check the source to verify that this is the current version before use.

4.5 Quality conformance inspection.

4.5.1 Procurement conditions. The following inspection plans shall apply where the assembly is procured for use as a:

Use	Plan
a. Logistics spare assembly apart from the next higher assembly	A and C
b. Component assembly of the next higher assembly	B and C

4.5.2 Inspection plan A.

4.5.2.1 General sample and tests. One assembly, as a control sample, shall be selected at random by the Government representative from each 100 assemblies produced and shall be subjected to all the tests in table IV.

4.5.2.2 Environmental sample and tests. Three assemblies, as a control sample, shall be selected at random by the Governmwnt representative from each 50 assemblies produced or from each month's production, whichever occurs first. All the tests in table V shall be applied separately to each assembly in the sample.

4.5.2.3 Acceptance. Where any one assembly of either sample fails to meet any specified requirement, the lot shall be rejected. Rejected lots shall be subjected to the provisions of MIL-STD-105.

4.5.3 Inspection plan B.

4.5.3.1 Sample and tests. The sample shall be selected by the Government representative in accordance with the provisions of MIL-STD-105. All the tests in table IV and only the test of item 307 in table V shall be applied.

4.5.3.2 Acceptance. Acceptance and rejection shall be in accordance with MIL-STD-105.

4.5.4 Inspection plan C.

4.5.4.1 Sample and tests. Each assembly in every lot shall be subjected to all the tests in table III and shall be examined visually for completeness, improper assembly and evidence of poor workmanship.

4.5.4.2 Acceptance. Where any one assembly fails to meet any specified requirement, the defective assembly shall be removed from the lot and resubmitted only after all defects have been corrected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:28Z
Check the source to verify that this is the current version before use.

4.5.5 Inspection equipment. Unless otherwise specified in the contract, the contractor shall supply, maintain and calibrate inspection equipment in accordance with the provisions of MIL-I-45607 and MIL-STD-45662.

4.5.6 Packaging inspection. The sampling and inspection of the preservation - packaging, packing and container marking shall be in accordance with the provisions of packaging data sheet P11732350.

4.6 Methods of inspection

4.6.1 Test equipment. Unless the assembly is procured apart from the next higher assembly, the test equipment in table VI shall be used where specified to perform required tests.

TABLE VI. Test equipment.

Item	Part Number	Nomenclature
1	11750202	Automatic Card and Unit, Test Set
2	11750354	Adapter, Computer Board - A3
3	11821457	Fixture Shock, Vertical and Horizontal Plane
4	11821458	Shock Adapter, Plate Vib Horz-Vert Plane/Shock -Z
5	11821500	Vibration/Shock, Test Fixture - Computer
6	11821520	Automatic Test Assembly Central Processor Unit
7	11821854	Test Fixture - Auto Card, Test set, Computer FAT

4.6.2 Performance tests.

4.6.2.1 D SERVO MOTOR output. Use items 1, 2 and 6 of table VI or suitable equipment to verify the requirements of 3.3.1.

4.6.2.2 E MOTOR 1 and E MOTOR 2 outputs. Use items 1, 2 and 6 of table VI or suitable equipment to verify the requirements of 3.3.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:28Z
Check the source to verify that this is the current version before use.

4.6.3 Environmental tests.

4.6.3.1 Shock

4.6.3.1.1 Basic design. Use items 3, 4 and 5 of table VI or mount the assembly in a suitable shock fixture and subject the assembly to the shock test specified in MIL-STD-810, method 516.2, procedure I, figure 516.2-2. Apply three half sine wave shock pulses in each direction along the three axes. Peak amplitude shall be 40 ± 4 g with a time duration of 18 ± 3 msec measured at the 10 percent amplitude points. At the conclusion of this test, subject the assembly to the performance tests of table III.

4.6.3.1.2 High intensity. Use items 3, 4 and 5 of table VI or mount the assembly in a suitable shock fixture and subject the assembly to the shock test specified in MIL-STD-810, method 516.2, procedure IV, figure 516.2-2. Apply three half sine wave shock pulses in each direction along the three axes. Peak amplitude shall be 100.0 ± 10.0 g with a time duration of 1.5 ± 0.2 msec measured at the 10 percent amplitude points. At the conclusion of this test, subject the assembly to the performance tests of table III.

4.6.3.2 Vibration. Vibration A as specified in 4.6.3.2.1 shall be conducted only for first article inspection. Vibration B as specified in 4.6.3.2.2 shall be conducted only for quality conformance inspection.

4.6.3.2.1 Vibration A. Use items 4 and 5 of table VI or mount the assembly in a suitable vibration fixture and subject the assembly to the vibration test specified in MIL-STD-810, method 514.2, procedure VIII, except that the test level shall be the vibration curve shown in figure 2 and duration of exposure shall be not less than 80 minutes per axis. At the conclusion of this test, subject the assembly to the performance tests of table III.

4.6.3.2.2 Vibration B. This test shall be conducted as in 4.6.3.2.1 except that the period of vibration shall be not less than 15 minutes in each axis.

4.6.3.3 High temperature. Use item 7 of table VI or suitable equipment and subject the assembly to the high temperature test specified in MIL-STD-810, method 501.1, procedure I. The highest operating temperature shall be 140°F. The performance tests of table III shall be applied.

4.6.3.4 Low temperature. Use item 7 of table VI or suitable equipment and subject the assembly to the low temperature test specified in MIL-STD-810, method 502.1, procedure I. The storage temperature shall be -65°F. The lowest operating temperature shall be -25°F. The performance tests of table III shall be applied.

4.6.3.5 Temperature cycling. With power and loads of table I applied, subject the assembly to the temperature profile specified in figure 3. At the conclusion of this test, subject the assembly to the performance tests of table III.

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:28Z
Check the source to verify that this is the current version before use.

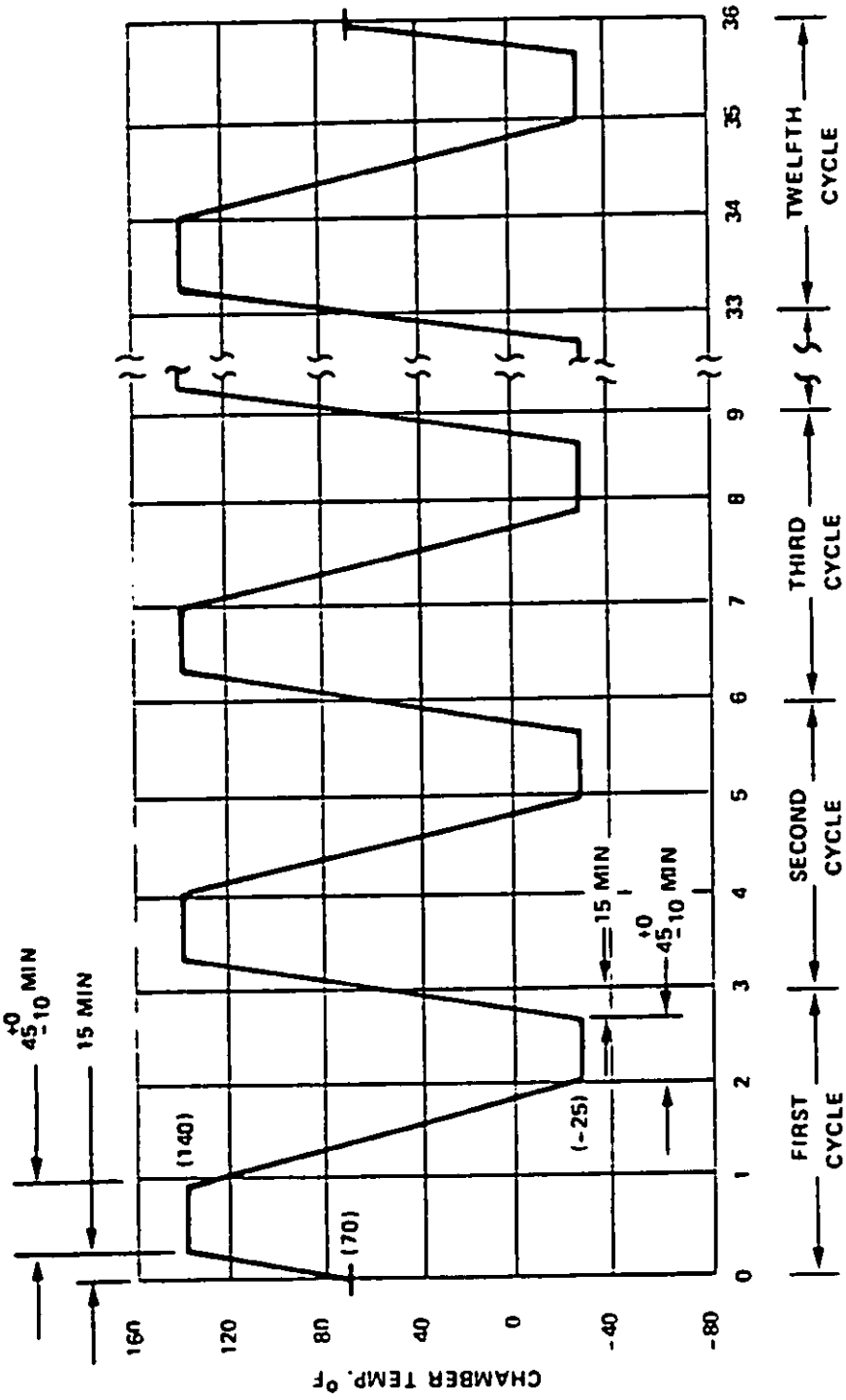


FIGURE 3. Temperature cycling.

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:28Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with packaging data sheet P11732350. The level of protection shall be as specified in the procurement document. (See 4.5.6)

6. NOTES

6.1 Intended use. The Reticle Projector and Elevation Servo Drives circuit card assembly is used in the M21 Ballistic Computer Unit.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Special tests and examination as required, the type of inspection required, and the activity responsible for testing and its facility.

c. Applicable packaging data sheet number (see 5.1).

d. Selection of applicable levels of preservation, packaging and packing.

e. Applicable stock number.

f. First article inspection requirements.

6.3 Previous revisions of MIL-C-48470B was for the US Army Armament Research and Development Command's internal use only, and were not forwarded to the Naval Publications and Form Center for publication.

Custodian:
Army-AR

Preparing Activity:
Army-AR
Project 1220-A276

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:28Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.
NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-C-48470B CIRCUIT CARD ASSEMBLY: 11732350 RETICLE PROJECTOR AND ELEVATION SERVO DRIVE

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:28Z
Check the source to verify that this is the current version before use.